

DYLAN FERREIRA

DATA ENGINEER | AZURE & MICROSOFT FABRIC | SQL | PYTHON | ETL/ELT

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Portfolio: <https://similur.github.io/Portfolio/>

Summary

Data engineer and database administrator with 7+ years of IT experience, including nearly 5 years in database administration, Azure data platforms, and enterprise integrations. Built ETL/ELT pipelines with Azure Data Factory, Microsoft Fabric, Python, and PySpark to automate manual processing and support reliable reporting across 26 campuses.

Technical skills

Data platforms: Microsoft Fabric, Azure Data Factory (ADF), ADLS Gen2, Fabric Lakehouse, Delta Lake, SQL Server, Azure SQL, PostgreSQL

Languages and integration: Python, PySpark, SQL, T-SQL, PowerShell, SQLAlchemy, pyodbc, REST APIs, JSON, Azure Key Vault

Data engineering: ETL/ELT, data modeling and warehousing, medallion architecture, Bronze/Silver layers, incremental processing, schema management, data validation

Database operations: Query optimization, stored procedures, indexing, backup and recovery, security, access controls

Analytics tools: Power BI, Pecan AI, Azure Document Intelligence

Professional experience

Junior Database Administrator (Data Engineering & Integrations) | San Joaquin Valley College

01/2022 - 09/2026

- Built and maintained Azure Data Factory pipelines and dataflows and Microsoft Fabric Bronze/Silver layers supporting reporting across 26 campuses. Integrated Excel, JSON, REST APIs, Azure Storage, SQL Server, and Azure SQL into the data platform.
- Automated daily Excel imports with PySpark in Microsoft Fabric, reducing a roughly 2-hour manual process to 5 minutes total. Implemented month-to-date/year-to-date (MTD/YTD) reconciliation to handle overlapping records and preserve historical consistency.
- Helped move reporting workloads from transactional SQL Server databases into Azure Data Lake and Microsoft Fabric, separating operational processing from analytical workloads.
- Administered SQL Server, Azure SQL, and PostgreSQL; tuned queries, developed stored procedures, maintained indexes, and supported backup and recovery, security, and access controls.
- Resolved Delta Lake merge conflicts, schema drift, datatype issues, and downstream Power BI data availability problems. Built Instructure and Azure Document Intelligence integrations using Python, PowerShell, T-SQL, SQLAlchemy, pyodbc, REST APIs, and Azure Key Vault to automate data movement.
- Developed, trained, and deployed 4 predictive models for student retention and financial risk. Leadership shared predictions with operational teams to guide outreach aimed at reducing student drops and potential bad debt.

Selected projects

Microsoft Fabric Data Pipeline Modernization

- Verified automated loads against existing manual imports using Power BI reconciliation workflows to check data accuracy during the transition to automated processing.

Instructure Canvas Data 2 Integration

- Built Python synchronization scripts running daily in Microsoft Fabric to ingest Canvas LMS enrollment, course, submission, and user data for reporting and analytics.
- Validated data with SQL and troubleshoot long-running syncs, consistency issues, and integration failures, coordinating with vendor support when needed.

Academic & Financial Predictive Modeling

- Used Pecan AI to prepare and validate historical data, evaluate predictive features and model performance, and support outreach for academic risk, early dropout, repayment, and potential bad debt.

Additional experience

Help Desk Specialist, Tier 1 | San Joaquin Valley College | 09/2019 - 12/2021

Provided Tier 1 support across 26 campuses, including Active Directory/Azure AD administration, JAMF-managed MDM deployments, troubleshooting, and self-service documentation.

IT Field Technician | Mobile Technologies Inc. (MTI) | 05/2019 - 09/2019

Installed and serviced retail security displays, POS systems, and network devices; performed Cat-5/6 and PoE cabling, troubleshooting, and on-site repairs.

Computer Repair Technician | Ferreira PC Repair | 01/2018 - Present

Build custom PCs and troubleshoot hardware, operating systems, and networks for individual clients.

Education & certifications

Associate of Science in Information Technology | San Joaquin Valley College | 01/2017 - 10/2018

Microsoft Certified: Azure Fundamentals (AZ-900) | Microsoft Certified: Azure Data Fundamentals (DP-900)